IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc Control Systems

Corrected copy

GAME THEORY

Wednesday, 8 May 10:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : D. Angeli

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Information for candidates:

No special instructions for candidates.

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1. Take an $n \times m$ matrix S. Consider a zero sum matrix game in mixed strategies:

$$\underset{y \in \Lambda_n}{\text{Min}} \ \underset{z \in \Lambda_m}{\text{Max}} \ y^T S z \tag{1}$$

in which, for a given integer r, Σ_r denotes the set of probability mass functions

$$\Lambda_r := \{(p_1, \ldots p_r) \ : \ \sum_i p_i = 1 \text{ and } p_i \geq 0 \text{ for } i = 1, \ldots, r\}.$$

(a): Assume that S is the matrix

$$\begin{bmatrix} 6 & 5 & 7 & 0 \\ 5 & 7 & 0 & 8 \end{bmatrix}.$$

By graphical means, or otherwise, derive a Nash equilibrium $(\bar{y}, \bar{z})$ for the game. [12]

Evaluate the pay-off V. [2]

(b): Let $(\bar{y}, \bar{z})$ be a Nash equilibrium for (1), with pay-off V. Show that $(\bar{y}, \bar{z}, \bar{p} := -V, \bar{q} := V)$ is a solution to the optimization problem

$$(P) \begin{cases} \text{Minimize } p + q \\ \text{over scalars } p \text{ and } q \text{ and vectors } y \in \Lambda_n \text{ and } z \in \Lambda_m, \\ \text{subject to} \\ Sz \geq -p 1_n \text{ and } S^T y \leq q 1_m, \end{cases}$$

in which, for a given integer r, 1_r denotes the r dimensional vector $[1, \ldots, 1]^T$. [6]

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2. Consider the 2 stage zero sum game summarized by the two diagrams Fig. 1(a) and Fig. 1(b), involving players A and B, and also a third player N ('Nature'), who plays a fixed random strategy known to players A and B. Players A and B aim to minimize and maximize the pay-off, respectively.

A acts first and B acts second. There is a communication link between the players, providing information about A's strategy to B. Nature determines, with equal probability, whether, or not, there is a fault on the link, which prevents the information being transmitted. Thus,

Nature chooses Scenario 1 ('no fault') w.p. 1/2 and Scenario 2 ('fault') w.p. 1/2, where

Scenario 1 (no fault): B knows A's action as in Fig. 2(a)

Scenario 2 (fault): B does not know A's action, as in Fig. 2(b)

Reduce the game to normal form, according to the following method.

Step 1: Identify the 8 possible strategies, $(B_1)\ldots,(B_8)$ available to B, depending on the actions of A and Nature. [10]

Hint: A representative strategy for B, (B_1) say, is

$$(B_1): B \text{ chooses } \begin{cases} L & \text{if } A \text{ chooses } L, \text{ and Nature chooses 'no fault'} \\ R & \text{if } A \text{ chooses } R, \text{ and Nature chooses 'no fault'} \\ L & \text{if Nature chooses 'fault'} \end{cases}$$

Step 2: Calculate the two 2×8 pay-off matrices, relating to the scenarios 'no fault' and 'fault' and average them according to the probabilities of occurence. [6]

Identify any Nash equilibria, and state whether they are admissible. [4]

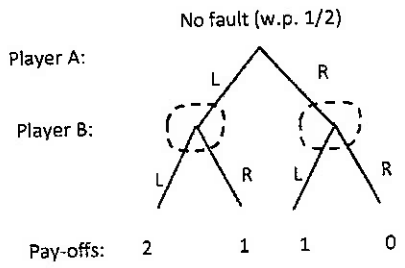


Fig. 2(a)

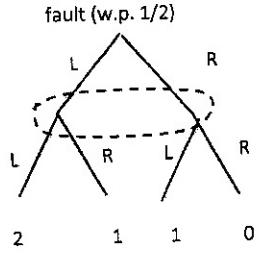


Fig. 2(b)

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3: Consider a game with two players, A and B. Both players aim to minimize their pay-offs. The pay-offs of players A and B are

$$L^A(a,b) := -ab + \tfrac{1}{2}a^2$$

$$L^B(a,b) := \begin{cases} -a^2b + \frac{1}{2}b^2 & \text{if } a \leq 2 \\ -4b + \frac{1}{2}b^2 & \text{if } a > 2\,, \end{cases}$$

respectively. The policy spaces are $\mathcal{A} = \{a : -\infty < a < \infty\}$ and $\mathcal{B} = \{b : -\infty < b < \infty\}$.

Sketch the response curves for players A and B. [4]

Determine all the Nash equilibria, and the pay-offs associated with each equilibrium. [6]

Which is the 'best' equilibrium and in what sense? [2]

Now consider a Stackelberg game for the two players, with the same pay-offs. Assume that A is the leader.

Find a function $\psi(a)$ such that, if $\bar{a}$ is a Stackelberg minimizer, then

$$\psi(\bar{a}) = \min_{a \in \mathcal{A}} \psi(a)\,.$$

[4]

Hence determine any Stackelberg minimizers $\bar{a}$ for A, and the optimal strategies for the B player when A chooses $\bar{a}$. [4]

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4. 3 isolated people, X_1, X_2, X_3, live on an island, when a volcano erupts. One person needs to sail to the mainland, to get help.

Because different people have different distances to sail, the pay-off for X_i (which X_i aims to maximize) depends on i. It is:

Satisfaction Level	Scenario
1	Someone (but not X_i) sails to the mainland
$1 - c_i$	X_i sails to the mainland
-1	no one sails to the mainland

(X_i gets high satisfaction (pay-off = 1) if someone sails, but her satisfaction is reduced by an inconvenience factor c_i, if she sails herself. She gets low satisfaction (pay-off = -1) is no one sails.)

Here c_1, c_2, c_3 (the inconvenience factors of X_1, X_2, X_3 respectively) are numbers satisfying

$$0 < c_1 < 1,\ 0 < c_2 < 1,\ 0 < c_3 < 1\,.$$

For $i = 1, 2, 3$, X_i must choose a mixed strategy independent of the other players' strategies.

Suppose that, for each i, X_i chooses a mixed strategy

'not sail' w.p. $(1 - \lambda_i)$, 'sail' w.p. λ_i

while, for $j \neq i$, X_j chooses

not sail' w.p. $(1 - \bar{\lambda}_j)$, 'sail' w.p. $\bar{\lambda}_j$.

Derive the expected cost for X_i, $i = 1, 2, 3$. [10]

Hence, or otherwise, derive three equations for $\bar{\lambda}_1, \bar{\lambda}_2$ and $\bar{\lambda}_3$ such that

$$\Big(((1 - \bar{\lambda}_1), \lambda_1), ((1 - \bar{\lambda}_2), \lambda_2), ((1 - \bar{\lambda}_3), \lambda_3)\Big)$$

is a Nash equilibrium in 'strict' mixed strategies, assumed to exist, for the three players. [4]

Now assume that

$$c_1 = c_2 = c_3 = c\,.$$

In the case $1/2 < c < 1$ solve the equations for the Nash equilibrium in strict mixed strategies. What is the probability that no one will sail? [4]

Is there is Nash equilibrium if $0 < c < 1/2$? [2]

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5. Consider the dynamic zero sum game

$$\begin{cases} \underset{\{u_t\}}{\text{Minimize}}\ \underset{\{v_t\}}{\text{Maximize}}\ \sum_{t=0}^{N-1} L(x_t, u_t, v_t) + g(x_N) \\ \text{over state feedback strategies } \{u_t\}_0^{N-1}, \{v_t\}_{t=0}^{N} \text{ and states} \{x_t\}_{t=0}^{N} \text{ such that} \\ x_{t+1} = f(x_t, u_t, v_t) \ \text{ for } t = 0, \ldots, N-1 \\ x_0 \text{ a given vector}. \end{cases}$$

(Here, $f(x,u,v)$, $L(x,u,v)$ and $g(x)$ are given functions of the vector variables (u,v,x).)

(A): Briefly describe the Dynamic Programming technique to determine a saddlepoint, in state feedback form, for this game. [5]

(B): Now assume that x, v and u are scalar variables and

$$f(x,u,v) = \left(x^3 + u + v\right)^{1/3},\ L(x,u,v) = u^2 - \gamma v^2 \text{ and } g(x) = x^6,$$

in which γ is a given positive number.

What is the Bellman equation for the value function $V_t(x)$, $t = 0, \ldots, N$? [3]

Assume that the value function is of the form

$$V_t = p_t x^6 \text{ for } t = 0, \ldots, N.$$

Assume also that the the zero-sum games appearing on the right side of the Bellman equation have a saddlepoint.

Derive recursive equations for the numbers $\{p_0, \ldots, p_N\}$ in these relations. [8]

Hence show that the Nash equilibrium strategies are of the form

$$u_t = k_t^{(u)} |x_t|^3 \text{ and } v_t = k_t^{(v)} |x_t|^3$$

and express the gains $k_t^{(u)}$ and $k_t^{(v)}$ in terms of the p_t's. [4]

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